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Crop Disease Detection and Advisory Systems

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Problem Statement

- **Lack of Early Disease Detection:** Farmers often fail to identify plant diseases at an early stage due to limited technical knowledge and visual similarity between diseases.

- **Overuse and Misuse of Pesticides:** Farmers often apply broad-spectrum pesticides unnecessarily, leading to increased production costs, soil degradation, and environmental pollution.

- **Lack of Real-Time Decision Support:** Farmers do not have access to instant, data-driven recommendations regarding disease severity, climate risk, and crop suitability during critical farming stages.

- **Absence of Crop Suitability Guidance:** Farmers often choose crops without analyzing soil pH and environmental compatibility. This reduces productivity and increases the risk of disease occurrence.

- **No Unified Platform for Image + Climate + Risk Analysis:** There is no integrated AI system that combines image-based disease detection with climate risk scoring and suitability indexing.

Dataset(s) details

- **Dataset:** PlantVillage Color Leaf Dataset
- **Training:** 80% of images
- **Testing:** 20%
- **Number of Classes:** 10 distinct disease

Link:- https://github.com/spMohanty/PlantVillage-Dataset/tree/master/raw/color

Literature Survey

1. "Plant leaf disease detection and classification using convolution neural networks model"- Tanko Daniel Salka, Marsyita Binti Hanafi, Sharifah M. Syed Ahmad Abdul Rahman, Dzarifah Binti Mohamed Zulperi, Zaid Omar (Springer Nature (2025))

- Methods: Comparative CNN analysis, transfer learning evaluation.
- Datasets: PlantVillage, ImageNet, custom field datasets.

Table 2 Taxonomy of plant leaf disease detection using CNNs

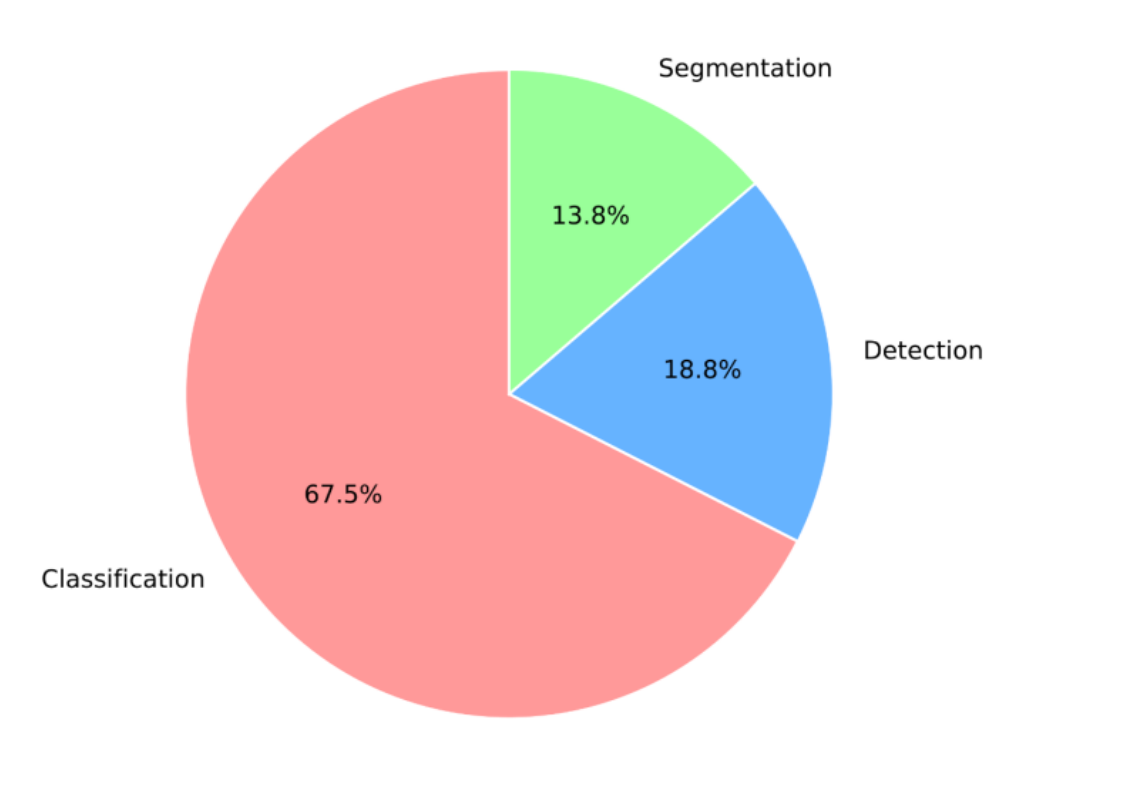
Category	Subcategory	Description
Input Data	RGB Images	Standard colour images of leaves showing visible disease symptoms.
	Multispectral/Hyperspectral	Multi-wavelength images for early detection, often from drones or cameras.
	Contextual Metadata	Supplementary data (e.g., weather, crop type) enhancing model accuracy.
Deep Learning Techniques	Segmentation Networks	
	- Fully Convolutional Networks (FCN)	Pixel-level classification of disease regions.
	- Mask R-CNN	Instance segmentation for precise lesion isolation.
	Detection Methods	
	- Two-Stage (R-CNN, SPP-Net)	Region proposal followed by classification, high accuracy but slower.
	- One-Stage (YOLO, SSD)	Direct prediction, fast but less precise for small targets.
	Classification Models	
	- VGG	High accuracy, computationally intensive.
	- ResNet	Robust with residual learning, top performer.
	- EfficientNet	Lightweight, efficient for resource-limited settings.
Datasets	Scope	
	- Single-Plant	Focused on one crop species.
	- Multi-Plant	Covers multiple species.
	Complexity	
	- Controlled Background	Simple settings, easier detection.
	- Natural Settings	Real-world complexity is challenging for models.
	Size	
	- Small	Limited images, constrains training.
	- Large	Extensive images, supports robust models.
4. Evaluation Metrics	Accuracy	Overall correctness of classification/detection.
	mAP	Mean Average Precision for detection performance.
	Precision/Recall	Balance of true positives vs. missed/falsely detected cases.
	Speed (FPS)	Frames per second for real-time capability.
5. Challenges	Data-Related	Small datasets, complex backgrounds limit performance.
	Technical	High computational cost, small lesion detection issues.
	Environmental	Lighting, occlusion, weather affect image quality.
6. Applications	Early Detection	Identifying diseases before visible symptoms escalate.
	Severity Estimation	Quantifying disease extent for management.
	Crop Management	Guiding farmer decisions with actionable insights.

Paper link: https://link.springer.com/article/10.1007/s10462-025-11234-6

Literature Survey Contd.

2. "A systematic review of deep learning techniques for plant diseases"- Ishak Pacal, Ismail Kunduracioglu, Mehmet Hakki Alma, Muhammet Deveci, Seifedine Kadry, Jan Nedoma, Vlastimil Slany,Radek Martinek (Springer Nature (2024))

- Methods: Classification, Detection, Segmentation model comparison.
- Datasets: PlantVillage, ImageNet, Multi-crop datasets.



Computer vision used in literature

Paper link: https://link.springer.com/article/10.1007/s10462-024-10944-7

Literature Survey Contd.

3. "Multi-Class Plant Leaf Disease Detection Using a Deep Convolutional Neural Network"- Shriya Jadhav, Anisha M. Lal (IGI Global (2022))

Models: Deep Convolutional Neural Network (CNN).

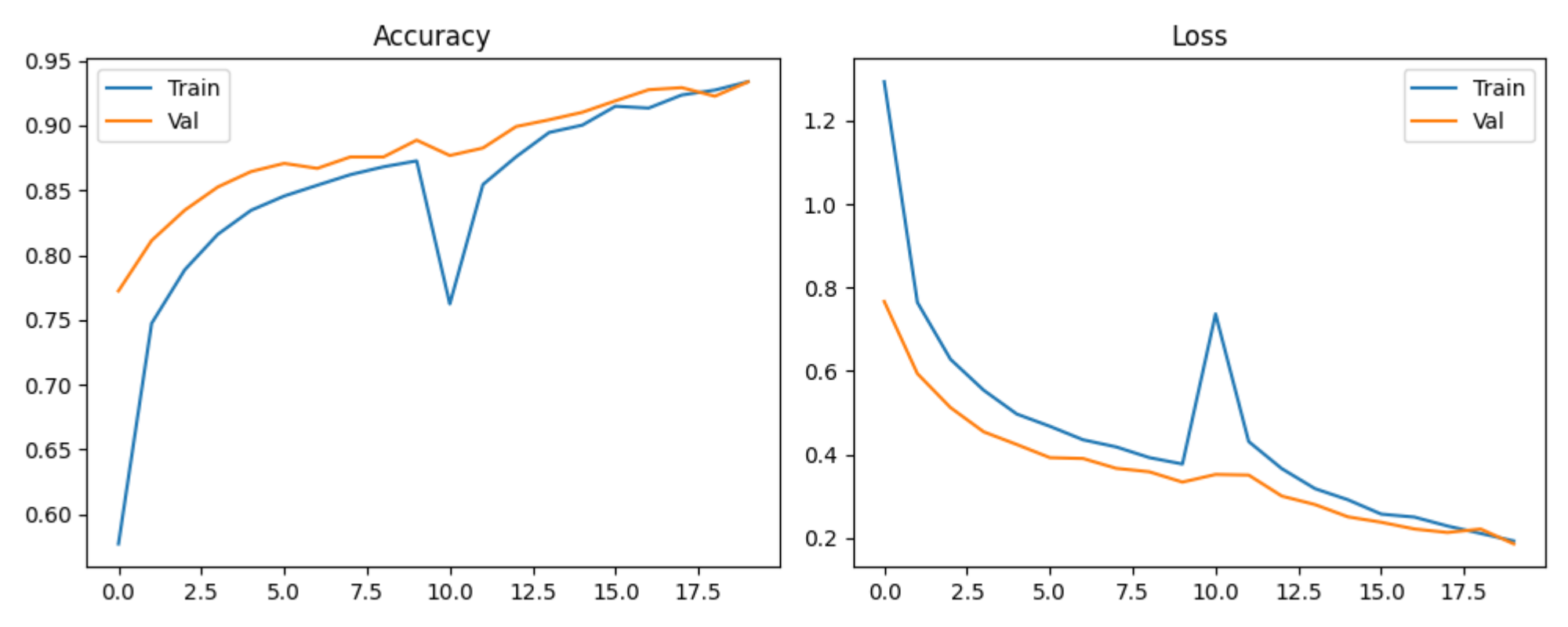
Datasets : PlantVillage, Field-collected leaf images.

Table 2. Configuration of DCNN

Layer	Sub-layer	Output Size	Stride	Feature Size
Input Layer	-	256 × 256 × 3	-	-
CNN Layer 1	Conv1	256 × 256 × 96	[1, 1]	6291456
	ReLU1	256 × 256 × 96	-	6291456
	MP1	128 × 128 × 96	[2, 2]	1572864
CNN Layer 2	Conv1	128 ×128 × 256	[1, 1]	4194304
	ReLU1	128 × 128 × 256	-	4194304
	MP1	64 × 64 × 256	[2, 2]	1048676
CNN Layer 3	Conv1	64 × 64 × 384	[1, 1]	1572864
	ReLU1	64 × 64 × 384	-	1572864
	MP1	32 × 32 × 384	[2, 2]	393216
Fully Connected Layer		13	-	

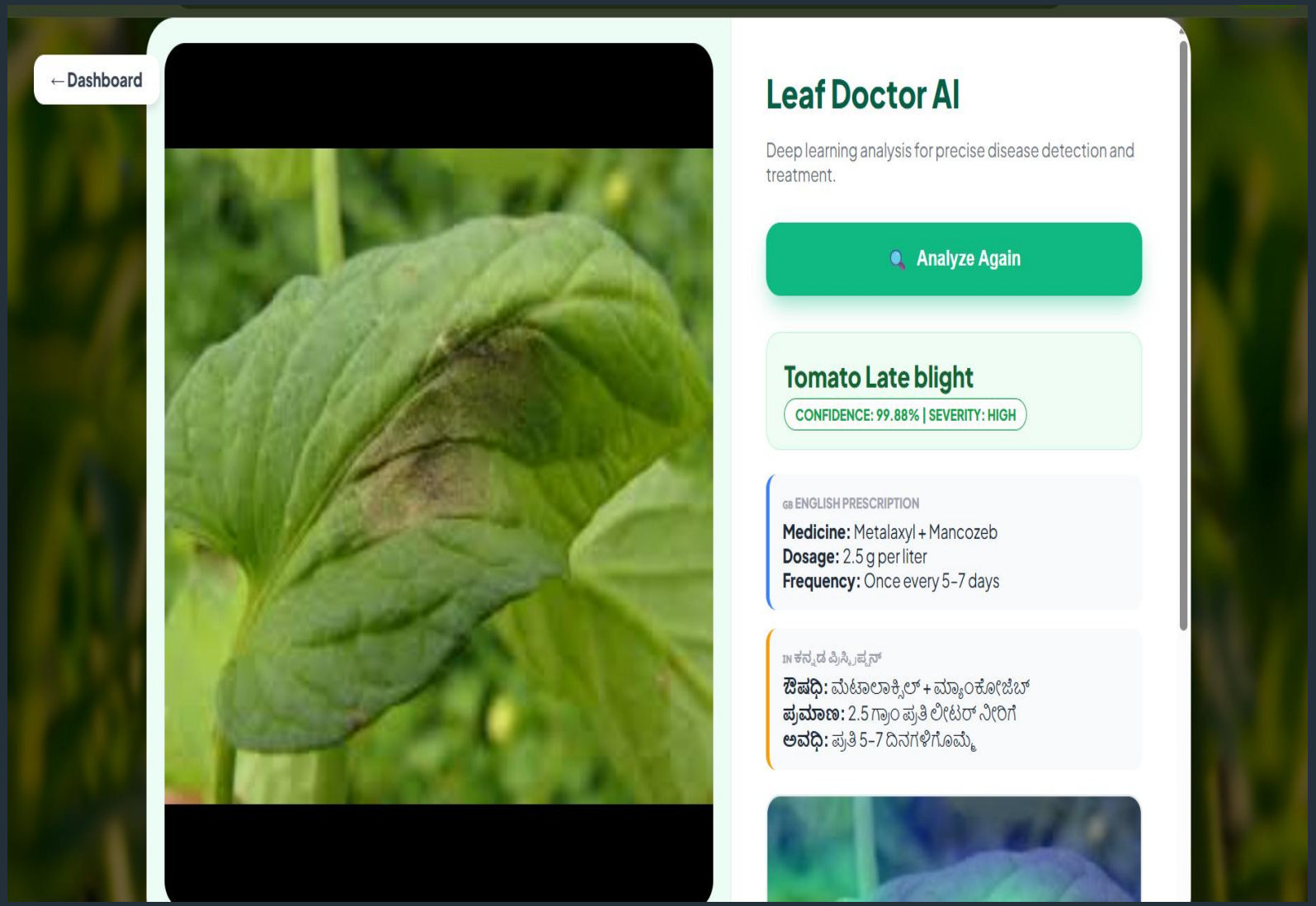
Paper link: https://www.igi-global.com/gateway/article/315126

Preliminary Results



https://arxiv.org/pdf/2102.00160

Preliminary Results



Further Plan

- Provide treatment recommendations and pesticide guidance in regional languages.
- Collect more field images under varying lighting and backgrounds.
- Include more crops and rare disease classes.
- Improve model generalization and reduce misclassification.
- Detect multiple disease spots in the same image.

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Thank you